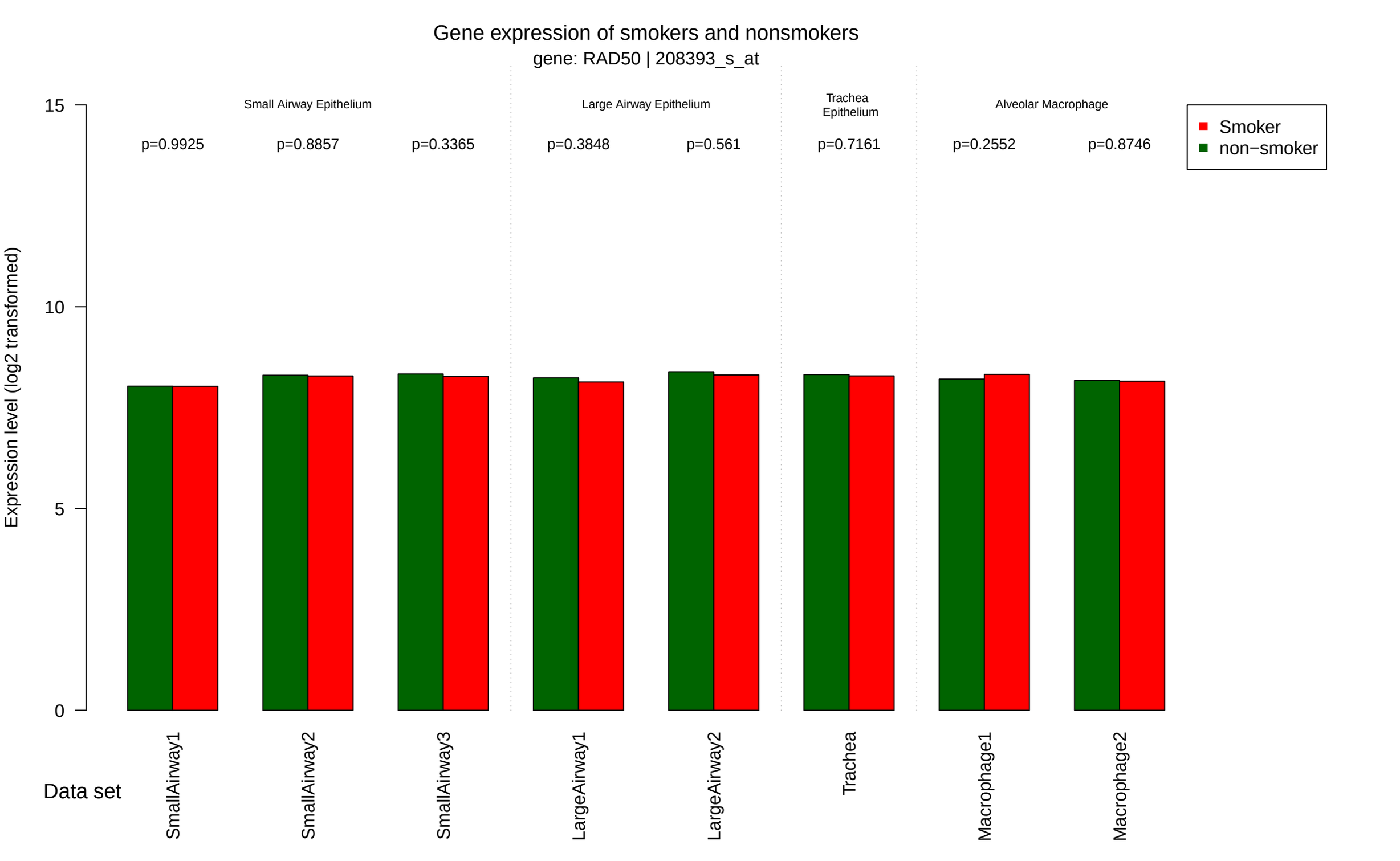
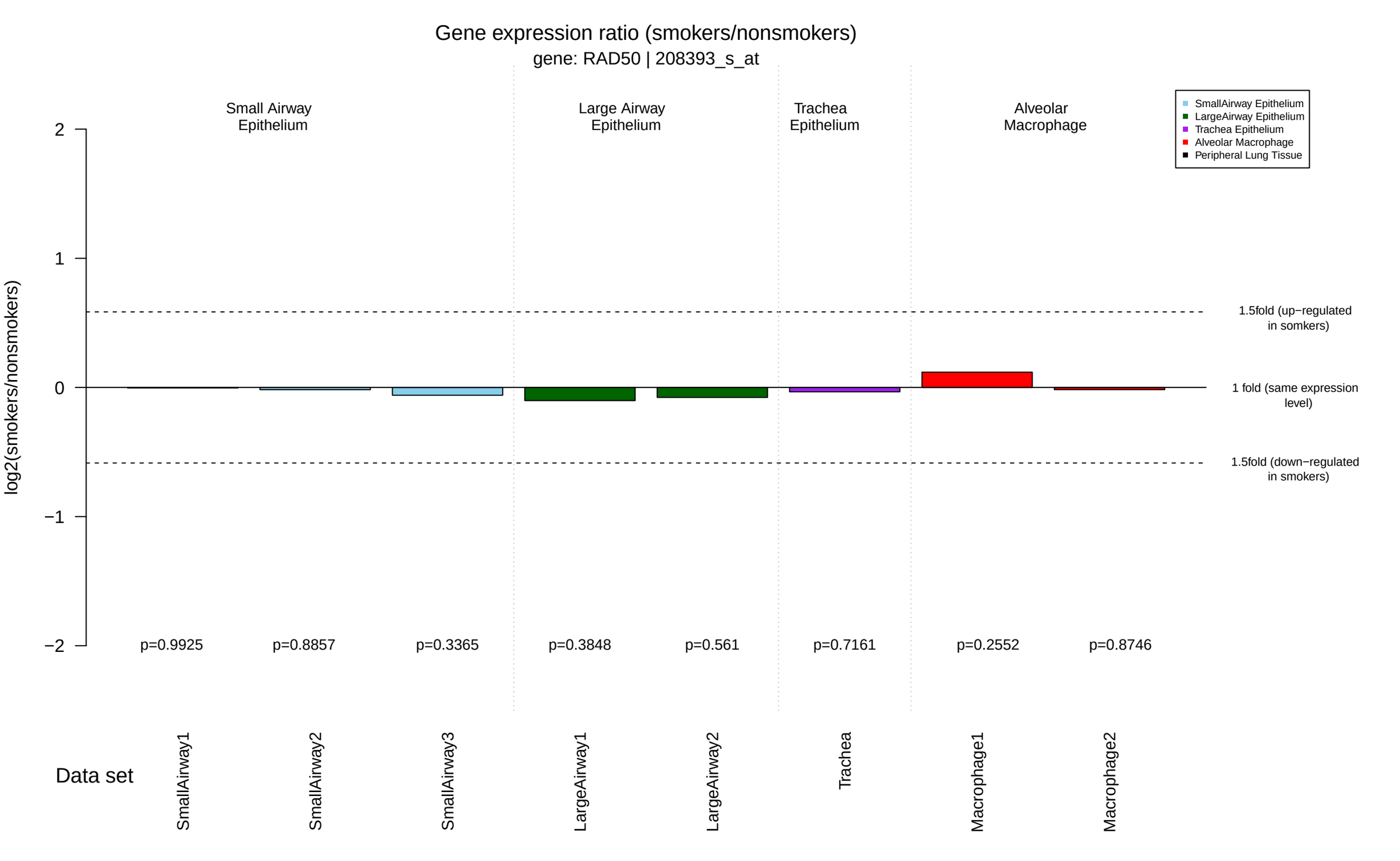
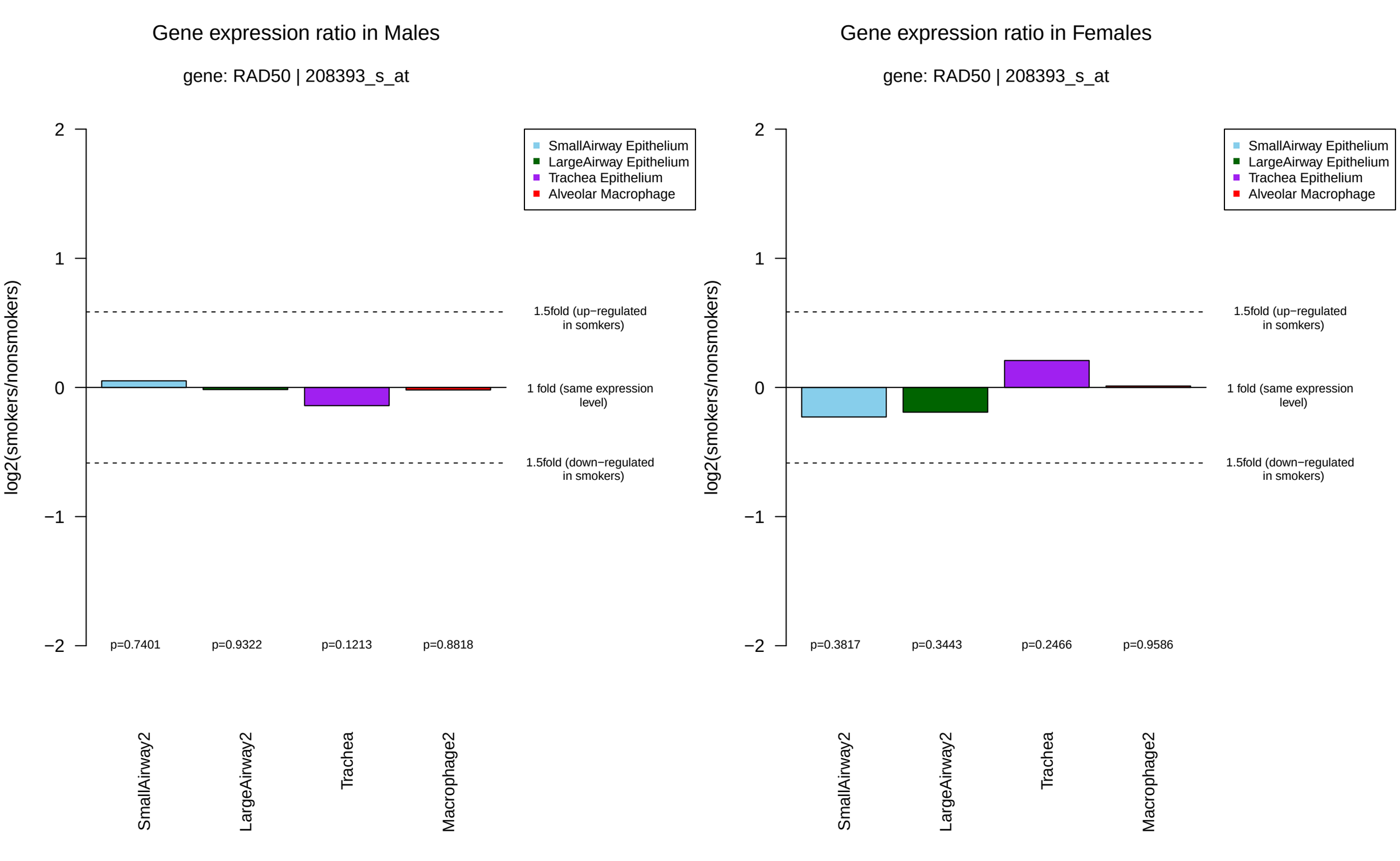
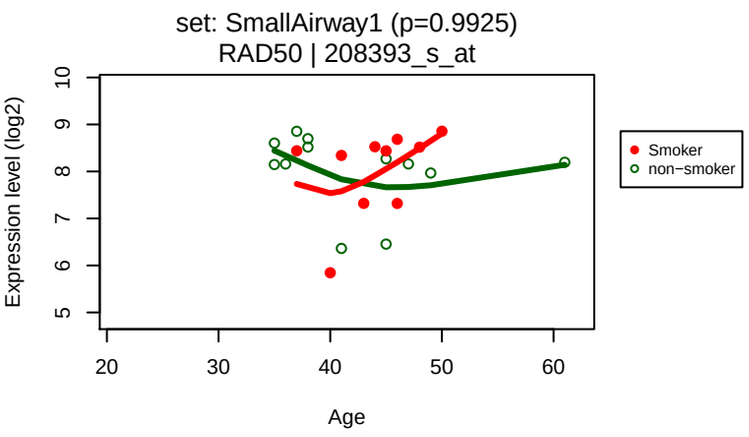
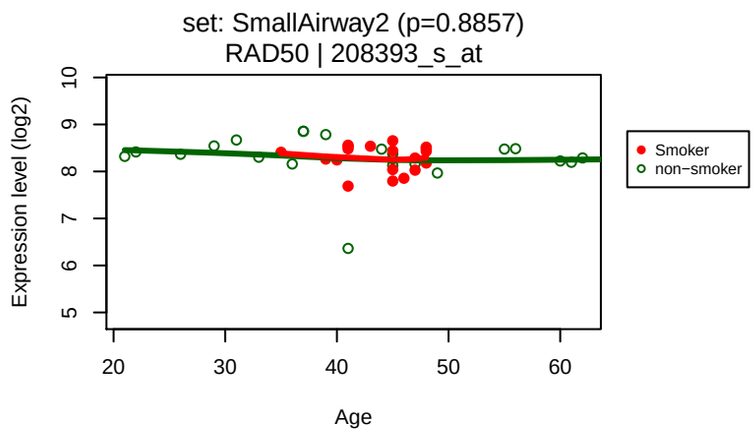
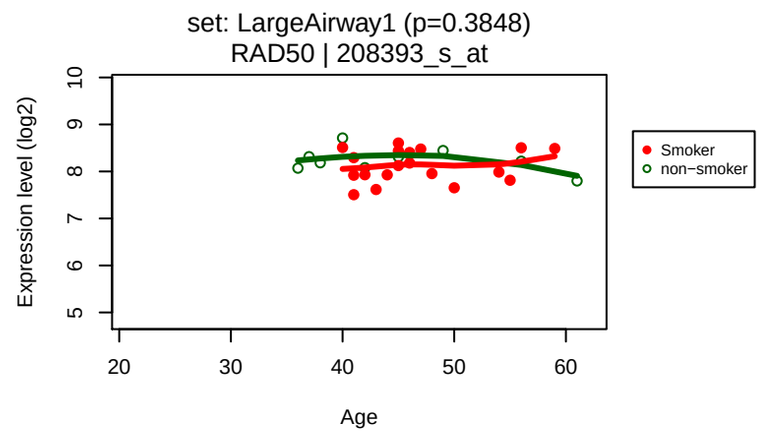

Gene expression of smokers and nonsmokers
gene: RAD50 | 208393_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9925
p=0.8857
p=0.3365
p=0.3848
p=0.561
p=0.7161
p=0.2552
p=0.8746
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

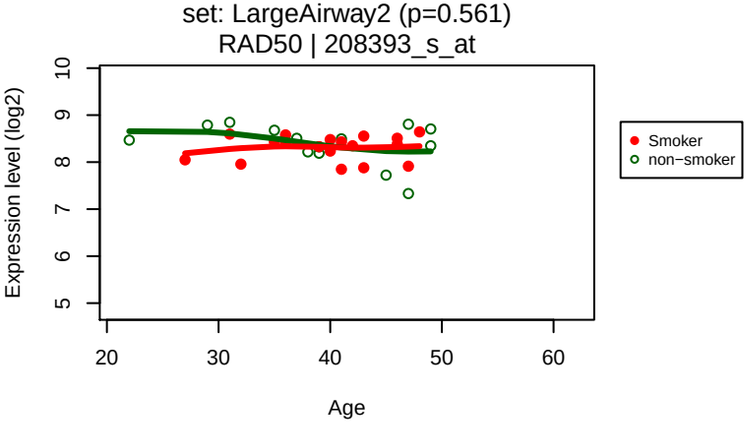

Gene expression ratio (smokers/nonsmokers)
gene: RAD50 | 208393_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9925
p=0.8857
p=0.3365
p=0.3848
p=0.561
p=0.7161
p=0.2552
p=0.8746
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

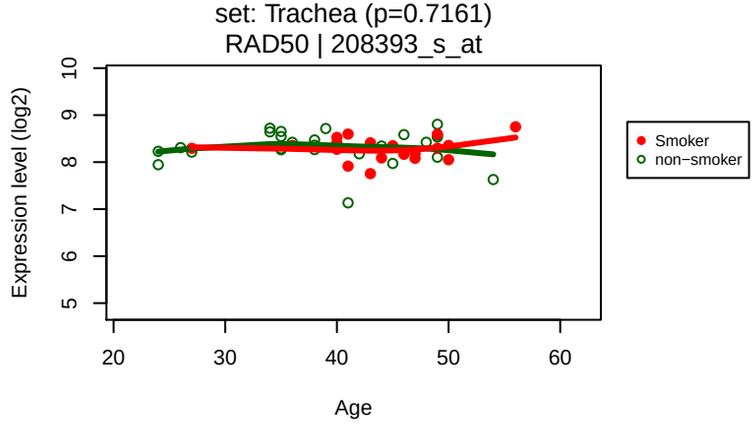

Gene expression ratio in Males
gene: RAD50 | 208393_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7401
p=0.9322
p=0.1213
p=0.8818
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAD50 | 208393_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3817
p=0.3443
p=0.2466
p=0.9586
SmallAirway2
LargeAirway2
Trachea
Macrophage2

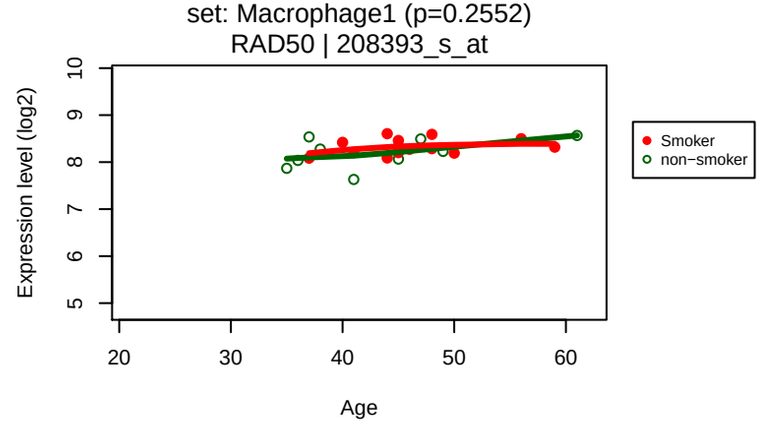

set: SmallAirway1 (p=0.9925)
RAD50 | 208393_s_at
Expression level (log2)
Age
Smoker
non−smoker

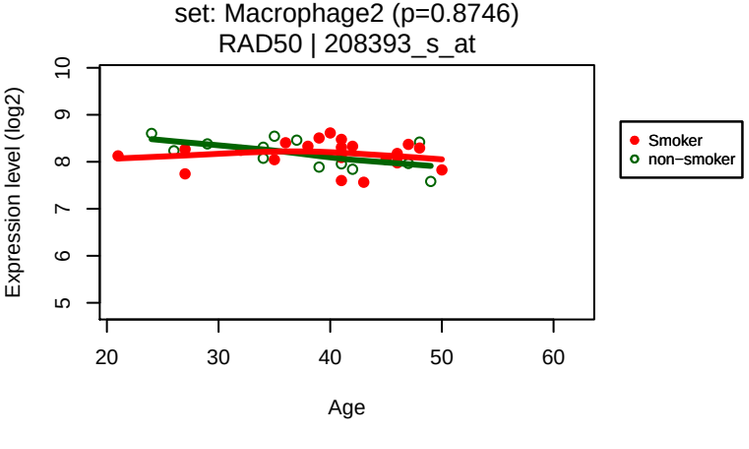

set: SmallAirway2 (p=0.8857)
RAD50 | 208393_s_at
Expression level (log2)
Age
Smoker
non−smoker

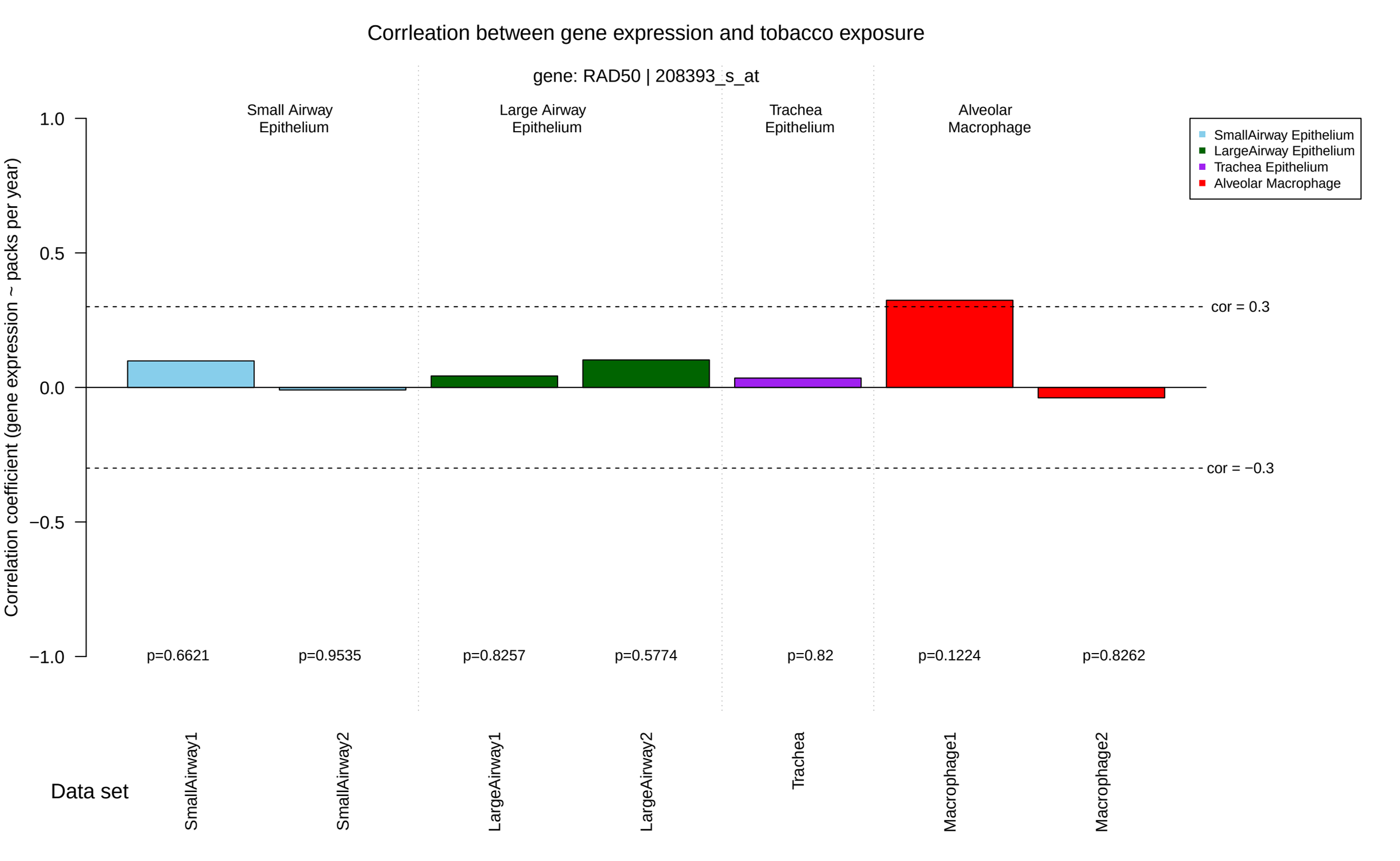

set: LargeAirway1 (p=0.3848)
RAD50 | 208393_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.561)
RAD50 | 208393_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7161)
RAD50 | 208393_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2552)
RAD50 | 208393_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8746)
RAD50 | 208393_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAD50 | 208393_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6621
p=0.9535
p=0.8257
p=0.5774
p=0.82
p=0.1224
p=0.8262
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2